Analysis of Professional Tennis Players and Match Data

1 Overview

This project introduces theory and applies statistical methods for studying information about professional tennis players and singles matches (between pairs of opponents) played, using data provided by the ATP (Association of Tennis Professionals) and WTA (Women's Tennis Association).

After a brief review of the literature, our first few sections introduce terminology on the tennis scoring system (point-game-set-match); we construct a Markov chain model and use curve-fitting as well as simulation to connect each layer of this hierarchical system with the next. Then we relate predicted likelihood of winning matches with the rankings of players to better understand the structure of this joint distribution while comparing AIC scores across different logistic regression models. And this evidence in our choice of the log-transform on rankings is additionally corroborated with bootstrapped confidence bands in a log-log GLM fit between the qualifying ITF (International Tennis Federation) rankings and the WTA Tour (which consists of the highest-level of tournaments in women's tennis). Next, we take a step back to critically question the consistency of rankings through the graphical theory of "domination cycles" which demonstrates that rankings are only approximations of skill level and not absolute in deciding overall win rates which are also influenced by idiosyncratic playstyles as suggested by real examples of 3-cycles in our domination digraph. Nonetheless, we wish to comprehend which variables can give us insight on win rates, and one such indicator is biological height of players that we inspect both longitudinally via hypothesis tests of difference from one generation to the successive generation, as well as via kernel smoothers to elucidate the complicated relationship between height and peak ranking achieved by each player. Another usage of smoothers brings out a side-by-side contrast between dropout or retirement ages on the men's ATP Tour and the women's WTA Tour, for which we confirm visually as well as perform formal tests of

statistical significance. Finally, we examine several pairs of covariates and the strength of nonlinearity in their bivariate relationships by way of difference between correlation matrices.

2 Review of Literature

One of the earlier studies on skill ratings and their relation to likelihood of one player winning over another is from Blackman and Casey in the article ”Development of a Rating System for All Tennis Players” [1], which along the way of modeling attributes about 1% of classification accuracy to court surface types, which is significant but not decisive between most pairs of players whose skill level varies to a greater extent (and this latter will in a later section be demonstrated through two-dimensional heatmaps).

In ”Performing best when it matters most: Evidence from professional tennis” [2], Díaz and Rogers decompose the hierarchical system of points, games, sets, and matches in a game-theoretic setting; they highlight the benefit of regarding longitudinal time series, which factor into parts of our work. Furthermore, the authors assert through citations of prior groundwork that the i.i.d. assumption on distribution of points between players throughout the course of a match is a suitable approximation for reality.

The application of Kendall’s tau in a correlation table by Gorgi, Koopman, and Lit’s “The analysis and forecasting of tennis matches by using a high dimensional dynamic model” [4] offers a way to assess performance in relation to surface types (clay, grass, hard court), and while we will include the same court surfaces (in addition to carpet) in bivariate analysis, we will utilize partial correlations and focus on nonlinearity instead by taking the difference between parametric linear coefficients and Spearman’s rank-based correlation. On top of the numerical summaries, where nonlinearity is strong we will report tail dependence if it exists as well.

Yue et. al in “A study of forecasting tennis matches via the Glicko model” [5] fit a curvilinear model of winning probability as a function of ranking and rating differences. However, we will show that the ratio of rankings (or difference in log-rankings) offers a better regression fit.

3 Theoretical Analysis of Game Win Rates within the Hierarchical Scoring System

Unlike many other sports such as basketball, football, and soccer, there are multiple layers of scoring in tennis; and after each layer, the score at the lower level of abstraction is reset. More concretely, from the highest level of abstraction downward we have:

1. A match consists of multiple sets, and the usual format is either best-of-three or best-of-five. As soon as one player wins a total of three sets in the best-of-five format, the match is over; and a player who accumulates two sets in the best-of-three format is declared the match winner.

2. A set is played until one player has won at least six games and the other player trails by at least two games (e.g. 6-4 or 7-5); or if both players have reached 6 games, then a 10-point tiebreak (the first player to reach 10 points wins) is held in place of the final game.

3. A game is played until one player has won four points and the other has at most 3 points; or if both players reach 3 points (this is called "deuce"), then points are played until one player has a 2-point lead; a one-point lead in favor of the server, the player who serves, is referred to as "advantage-in"/"ad-in"; and if the returner, the player who receives the serve, has a one-point lead it is called "advantage-out"/"ad-out".

Starting with the probability of winning a single point (and assuming that all such points are played out with the independent and identically distributed setting of a Bernoulli process [2]), we analyze winning probabilities for a game, set, and match from a bottom-up approach.

Let p, $0 < p < 1$, be the probability that player 1 wins a point, so that $q := 1 - p$ is the probability that player 2 wins a point. It makes sense to assume that p cannot be exactly 0 or 1, because there is always a non-zero possibility of uncertainty due to non-performance related interference (such as a sudden gust of wind blowing the tennis ball off course).

We model the process as a Markov chain. Let G_{ij} be the event that a game has reached a stage where player 1 (the "server") has won i points, and player 2 (the "returner") has won j points. Then G_{kl} is accessible from G_{ij} if and only if $i \leq k$ and $j \leq l$. All states where the game has ended are absorbing, and all other states are transient.

For all situations where i and j are both strictly less than 4, as well as in cases where one player wins all points, the probability that the server wins a game without reaching deuce can be calculated using the PMF of Binomial($i+j$, p), and we denote the colloquial jargon ("love": 0, "15": 1 point, "30": 2 points, "40": 3 points, "game": 4 points, "all": same number of points for both players) in each scenario below.

$P(G_{00}) = 1$: "love all"
$P(G_{01}) = q$: "love-15"
$P(G_{10}) = p$: "15-love"
$P(G_{02}) = qP(G_{01}) = q^2$: "love-30"
$P(G_{11}) = qP(G_{10}) + pP(G_{01}) = 2pq$: "15-all"
$P(G_{20}) = pP(G_{10}) = p^2$: "30-love"
$P(G_{03}) = qP(G_{02}) = q^2$: "love-40"
$P(G_{12}) = \binom{3}{1} p^1 q^2 = 3pq^2$: "15-30"
$P(G_{21}) = \binom{3}{2} p^2 q^1 = 3p^2 q$: "30-15"
$P(G_{30}) = p(G_{20}) = p^3$: "40-love"
$P(G_{04}) = q(G_{03}) = q^4$: server loses, returner wins in straight points ("breaks serve in golden game")
$P(G_{13}) = \binom{4}{1} p^1 q^3 = 4pq^3$: "15-40"
$P(G_{22}) = \binom{4}{2} p^2 q^2 = 6p^2 q^2$: "30-all"
$P(G_{31}) = \binom{4}{1} p^3 q^1 = 6p^3 q$: "40-15"
$P(G_{40}) = p(G_{30}) = p^4$: server wins in straight points ("holds serve in golden game"), returner loses
$P(G_{14}) = \binom{5}{1} p^1 q^4 - pP(G_{04}) = 5pq^4 - pq^4 = 4pq^4$: "15-game"
$P(G_{23}) = \binom{5}{2} p^2 q^3 = 10p^2 q^3$: "30-40"
$P(G_{32}) = \binom{5}{3} p^3 q^2 = 10p^3 q^2$: "40-30"
$P(G_{41}) = \binom{5}{1} p^4 q^1 - qP(G_{40}) = 5p^4 q = 4p^4 q$: "game-15"
$P(G_{24}) = \binom{6}{2} p^2 q^4 - qqqqpp - qqqpqp - qqpqqp - qpqqqp - pqqqqp = 10p^2 q^4$: "game-30"
$P(G_{42}) = \binom{6}{4} p^4 q^2 - ppppqq - pppqpq - ppqppq - pqpppq - qppppq = 10p^4 q^2$: "30-game"
$P(G_{33}) = \binom{6}{3} p^3 q^3 = 20p^3 q^3$: "deuce", or "40-all"

For events above where a player has won despite the other player having gained at least one point (which is the typical scenario for competitive games), the probability has to be determined ad-hoc by conditioning and we cannot simply apply the Binomial model because order does matter – if one player reaches 4 points, then the game finishes and the other player cannot gain any extra points, so sequences where the winner

gains 4 points and then the loser gains additional points are impossible by standard rules of tennis. For instance, in the G_{24} and G_{42} cases we must exclude the 5 outcomes each where the game finishes with the last point scored by the loser.

To be clear, G_{33} refers to the set containing all sequences of point play where deuce is reached. It includes "ad-in", "ad-out", and subsequent "deuces". Let D_k denote the event that deuce has occurred k times, AI_k that deuce has occurred k times followed by ad-in, and AO_k for D_k followed by ad-out.

$$\begin{aligned}
P(D_k) &= P(G_{33})(pq+qp)^k = (20p^3q^3)(pq+qp)^{k-1} = 2^{k+1}5(pq)^{k+2} \\
P(AI_k) &= pP(D_k) = 2^{k+1}5p^{k+3}q^{k+2} \\
P(AO_k) &= qP(D_k) = 2^{k+1}5p^{k+2}q^{k+3}
\end{aligned}$$

The probability that the server wins immediately following the k-th deuce and ad-in is then:

$$P(G_{3+2k,1+2k}) = pP(AI_k) = 2^{k+1}5p^{k+4}q^{k+2}$$

And similarly, the probability that the returner wins immediately following the k-th deuce and ad-out is then:

$$P(G_{1+2k,3+2k}) = qP(AO_k) = 2^{k+1}5p^{k+2}q^{k+4}$$

Hence, the overall probability $P(SD)$ that the server wins following any positive (integer) number of deuces is a geometric series:

$$P(\bigcup G_{3+2k,1+2k}) = \Sigma P(G_{3+2k,1+2k}) = 2^{k+1}5p^{k+4}q^{k+2}(1+2pq+(2pq)^2+(2pq)^3+\dots)$$

And similarly, the overall probability $P(RD)$ for the returner winning following deuce:

$$P(\bigcup G_{1+2k,3+2k}) = \Sigma P(G_{1+2k,3+2k}) = 2^{k+1}5p^{k+2}q^{k+4}(1+2pq+(2pq)^2+(2pq)^3+\dots)$$

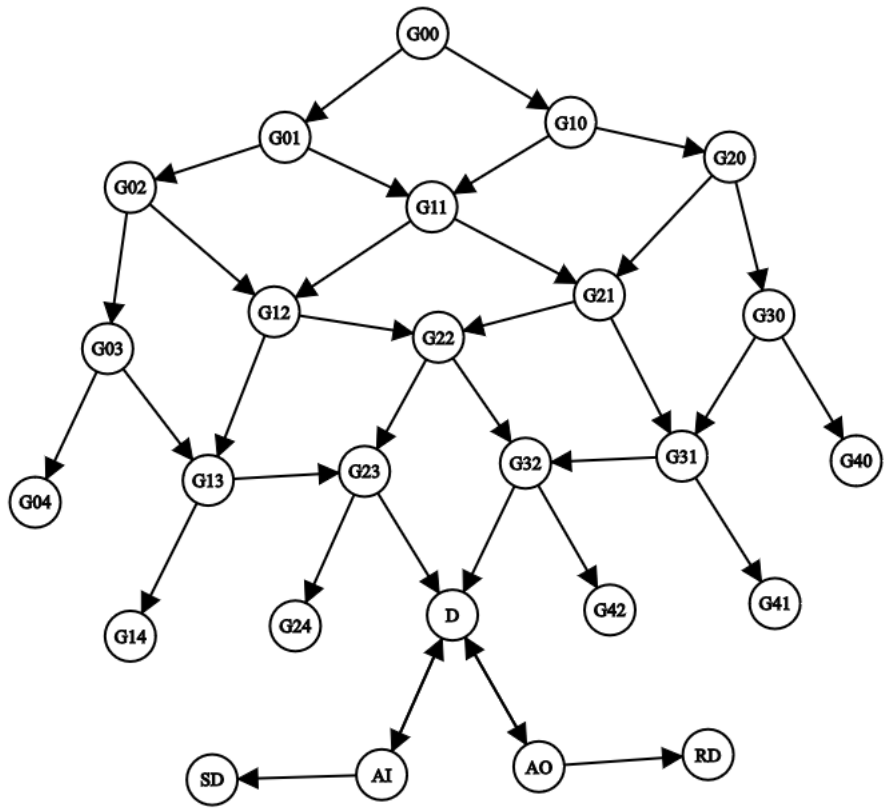
Thus, we can add up the disjoint events that the server wins to derive:

$$P(SG) = P(G_{40}) + P(G_{41}) + P(G_{42}) + P(SD)$$

And the event that the returner wins is the complement:

$$P(RG) = 1 - P(SG) = P(G_{04}) + P(G_{14}) + P(G_{24}) + P(RD)$$

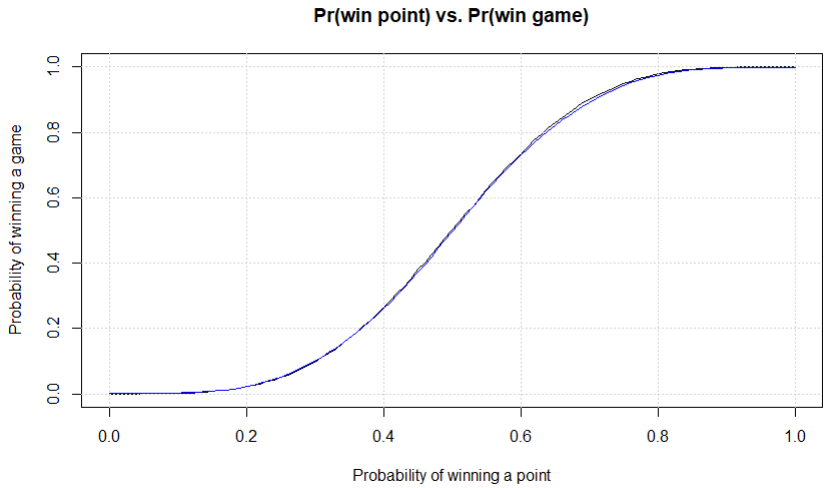
Although this notation makes it easier to calculate probabilities, if we treat each k-th deuce (for any positive integer k) all the same as G_{33} due to effective indistinguishability, and likewise for the ad-ins and ad-outs respectively, then it becomes easier to draw the corresponding Markov chain with finite states, and the steady-state probabilities remain unchanged through this modification:



4 Numerical Estimation of Win Rates within the Hierarchical Scoring System through Random Simulation

Having derived the probability of winning a game as a function of winning a point (to re-iterate, under the IID assumption as per Díaz and Rogers), we can likewise perform a similar analysis for probabilities of winning a set and winning a match. But it is not hard to realize that the expressions we obtain will become increasingly complicated in terms of the parameter p; rather, in this section we will see by exhibiting simulated curves how probability of winning a point, game, set, and match are numerically connected.

In the figure below, we show in the black curve a plot of numerically simulated game win rates as a function of the parameter p for probability of winning each point.

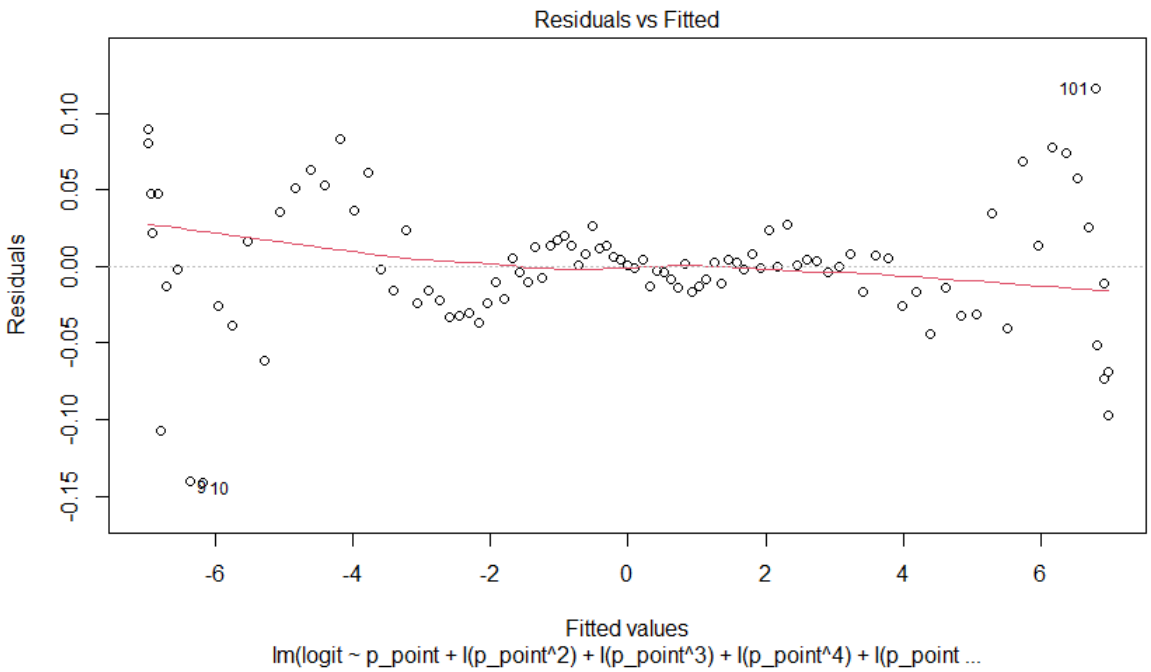


The moral of our story will be this: Even a slight difference in probability of winning each point leads to an exaggerated effect at the game-level. For instance, a player with just a 55% probability of winning a each point has a 62% likelihood of winning the game, 60% probability in points to 74% in games, and 70% probability in points corresponds to 90% win rate in games.

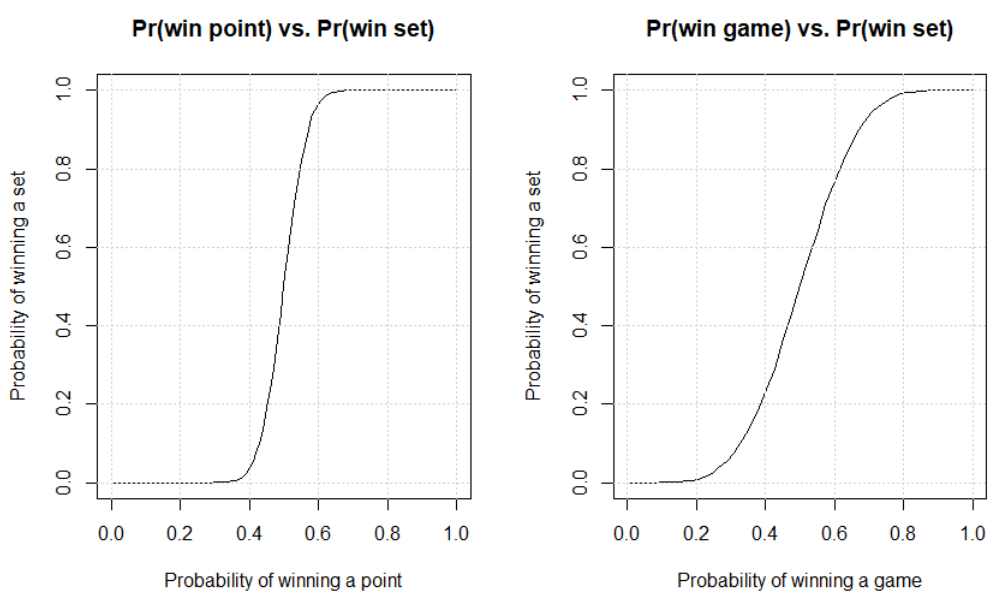
Since the curve has a sigmoidal shape, we can attempt to fit a logistic regression model (curve in blue) of the form:

$$logit(G) = \log(\frac{G}{1-G}) = -6.801 - 16.84p + 408.9p^2 - 2007p^3 + 4965p^4 - 6711p^5 + 4723p^6 - 1349p^7 + \varepsilon$$

Despite an adjusted R^2 of 99.99% and all t-values of the coefficients having absolute magnitude of at least 14 ($|t| > 14$) to suggest that we are not overfitting too many higher-order terms, the residual plot reveals that the quality of fit is closer near the center of the curve than near the endpoints which are less stable.

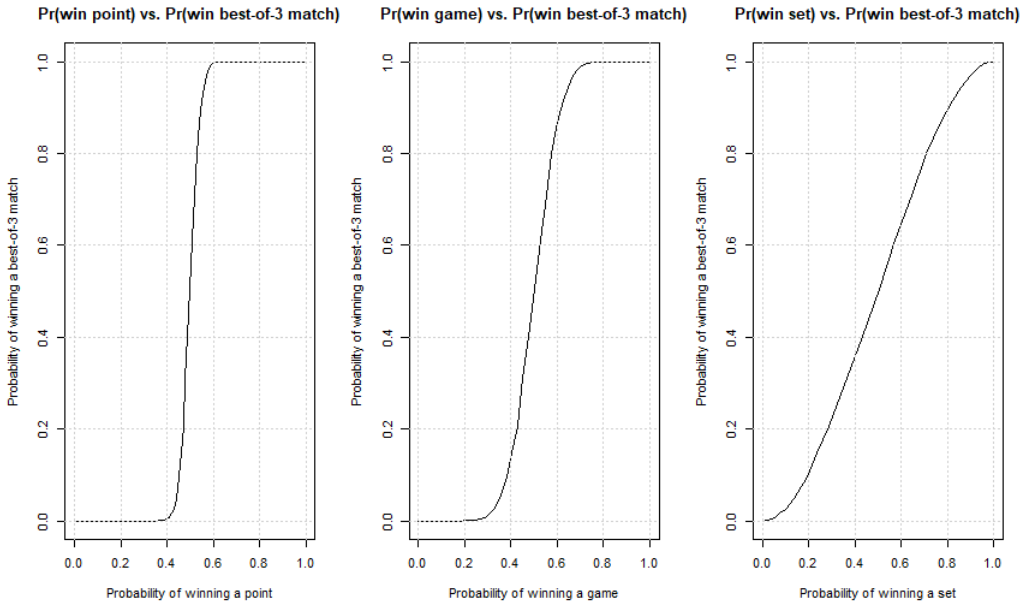


Officially, players alternate between service and return as well as switching sides of the court every game (to balance out exogenous exposure to sunlight and rain). And the player who serves in the first game is determined by a coin flip. As a simplifying approximation, we will use a single probability parameter of winning averaged over service and return games (which is justified by the fact that over the entirety of a match, about half of the games are serviced by one player and the other half by their opponent).

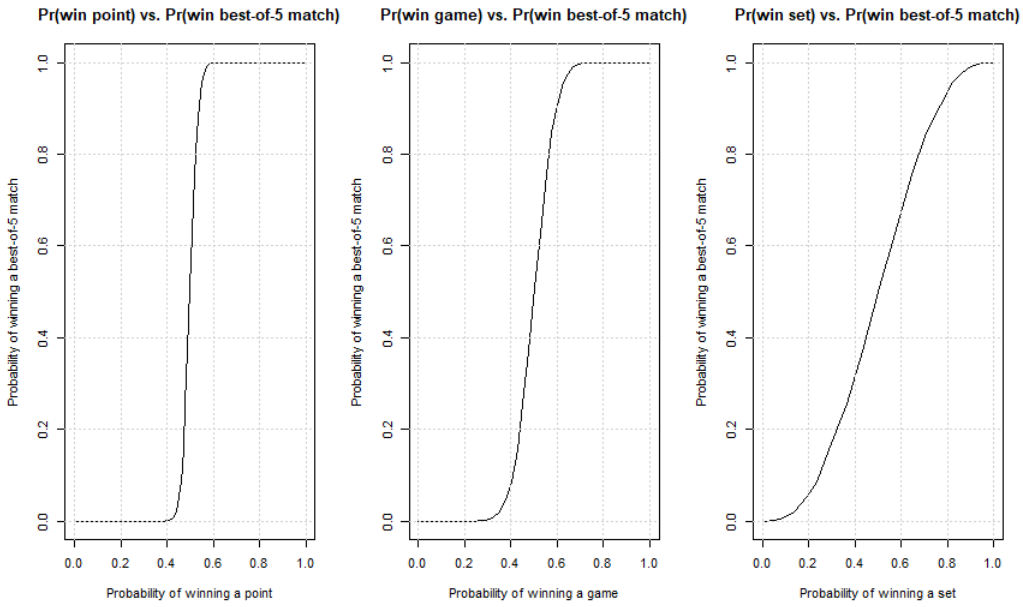


Evidently, each layer of abstraction in the scoring hierarchy exacerbates the skill gap to a greater extent; this is noticeable from the steepness (first difference) of the curve on the left. Continuing the way of example, a player with 55% probability of winning each point has a 82% likelihood of winning the set.

Since matches are generally played in a best-of-five format on the ATP Grand Slam tournaments and as best-of-three in other events, we look at the different simulation results separately.



In the best-of-three format, a 55% probability of winning a point translates to a 92% likelihood of winning the match.



In the best-of-five format, a 55% probability of winning a point translates to a whopping 96% likelihood of winning the match. This disparity is even more pronounced due to the higher number of sets played, and hereby a lower chance of the relatively weaker player winning sufficiently many early sets by luck.

In short, even a seemingly small difference in win rate can become more obvious by aggregation. This phenomenon is similar to the law of large numbers, except the statistical setting in our model is of a hierarchical type.

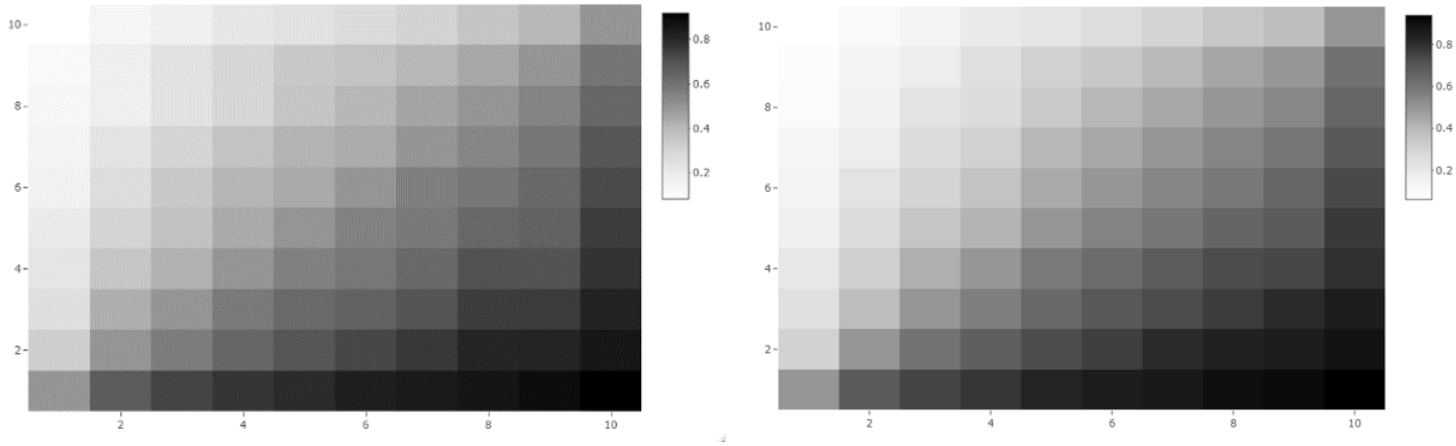
5 Modeling Match Win Rates as a Bivariate Surface over Rankings

In order to understand how close two opponents have to be in order for a match to be competitive, we can visualize with a 2D heatmap the likelihood of one player winning against another as a function of each player's current ranking. This higher-dimensional view is useful because without deciding a priori the functional form of transformation on the two players' rankings, we can take a non-parametric approach by plotting the empirical win rate using chromatic representation.

Since lower-ranked players (aka "journeymen" traditionally, or "journeywomen") tend to have more difficulty qualifying for many tournaments at the ATP and WTA tour level, and they are usually knocked out of tournaments in earlier rounds, most matches recorded are among top 100 players. Nevertheless, we argue that there is more benefit to be extracted from computing quantiles weighted by number of matches played because the ratio in rank between a rank 10 player and rank 100 is the same as between a rank 100 and rank 1000 player (so there is less meaning in separating out every 100 rankings between 100, 200, $\ldots$, 900, which holds true due to the distribution of quantiles for most distributions found in nature that aren't excessively negatively-skewed), and secondly the statistical promise of more stable estimates with consistent sample sizes is appealing for sake of providing us with reliability in comparing values (through shades of darkness) across the heatmap. With 142,097 matches on the ATP and 96,910 matches on the WTA with rankings of both players labeled, on average we will have on the order in magnitude of $10^3 = 1000$ for each of 100 grid cells in the map (although the partition will still be far from uniform, given that players with large differences in rank are less likely to play each other during later rounds of tournaments where the relatively far weaker contestant will seldom even make it that far). Standard errors can be bootstrapped within this framework, but given the forced sample sizes in the marginal distribution of rankings, we skip the redundant check.

Match frequency-weighted percentile	ATP ranking	WTA ranking
0%	1	1
10%	6	6
20%	13	12
30%	21	20
40%	32	30
50%	44	42
60%	58	57
70%	76	75
80%	99	100
90%	146	146
100%	2,101	1,503

In actuality, we find that at least 623 data points even in the sparsest decile-decile pair (where a top decile player loses a match against a bottom decile player) on both the ATP and the WTA datasets.



In the grayscale heatmap on the left generated for the ATP Tour, we have the win-rate for players from the 1st (top-ranked decile) to 10th (bottom-ranked decile) on the vertical axis versus players on the horizontal axis; and on the right is for the WTA Tour. For instance, an ATP player in the 1st decile (ranked in the top 6) has an 80% win rate against a player in the 5th decile (ranked between 44 to 58), and likewise the win rate is 83% for corresponding deciles on the WTA. Due to the similarity in the heatmaps, it is difficult to perceive much difference, and in fact the largest difference in magnitude is 6% for the 3rd

decile versus 9th decile win rate, of 76% on the ATP as contrasted to 82% on the WTA. Moreover, the entire upper-right triangle of the ATP matrix of win rates is lesser in value than the upper-right triangle of the WTA matrix, which implies that stronger WTA players tend to have a surer chance of beating a lower-ranked opponent than on the ATP tour; match results are more predictable from rankings in this sense. A passing remark is that these matrices as shown in the heatmaps are symmetric (trivially by definition, due to every match being a fixed-sum game with exactly one winner and one loser).

Now, turning to a (simple) logistic regression problem, we choose between five models for predicting likelihood of winning a match from a function $f : R^2 \to R$ of the two players' rankings, where x_1 and x_2 are the rankings of the two players:

1. Difference in rankings: $f(x_1, x_2) = x_1 - x_2$
   In this transformation, the difference between rankings 1 and 101 is treated as equal to the difference between rankings 900 and 1000, for instance.

2. Ratio of rankings: $f(x_1, x_2) = \frac{x_1}{x_2}$
   In this transformation, the ratio between rankings 1 and 10 is equal to the ratio between rankings 100 and 1000, for instance.

3. Logarithm of ratio in rankings: $f(x_1, x_2) = \log(\frac{x_1}{x_2}) = \log(x_1) - \log(x_2)$
   This is much the same as the second case, except we take a natural log transform for a non-linear fit.

4. Square root of ratio in rankings: $f(x_1, x_2) = \sqrt{\frac{x_1}{x_2}}$

5. Square of the ratio in rankings: $f(x_1, x_2) = (\frac{x_1}{x_2})^2$

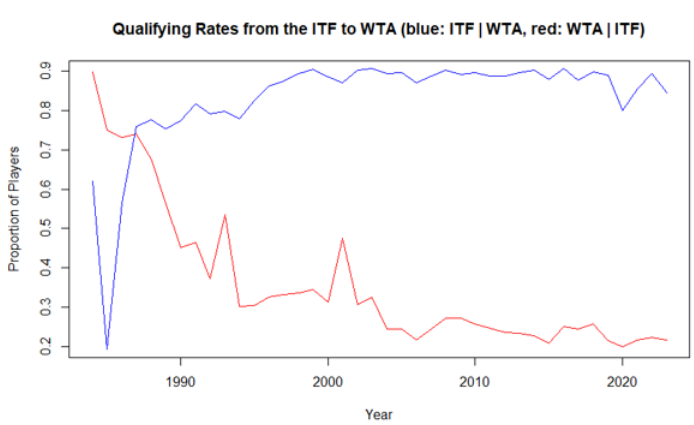
To select a model, we rely on AIC scores. For each model type, the WTA model had a lower AIC than the ATP model from the same row, which implies the same conclusion as our non-parametric methods showed that binary classification on the WTA Tour is more accurately decided by rankings than for the ATP Tour. And in both cases, the logarithm of ratios between the players rankings (or difference between log rankings) yielded the best predictor among the variety of model types considered. As a side comment, there is no need to evaluate these AICs out-of-sample in this context because first, we have more than plenty degrees of freedom; second, AIC already coincides with leave-one-out cross-validation for generalized linear models; and third, we are not trying to conduct statistical inference on the AIC scores; nevertheless, we could bootstrap on a hold-out test set if that were our intention as for other applications.

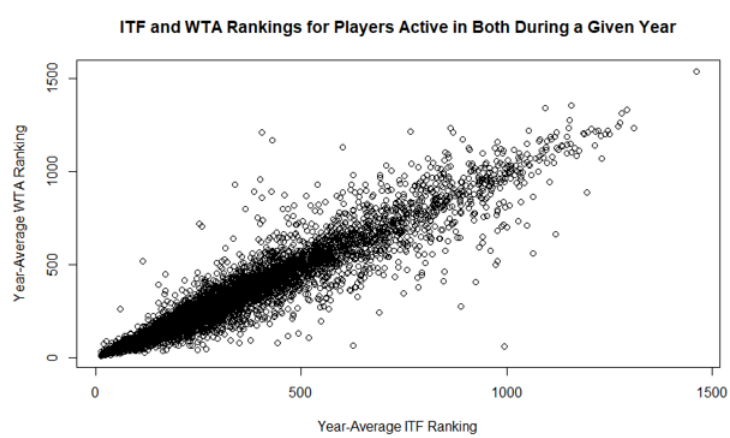
Model type	AIC on ATP data features	AIC on WTA data features
#1: difference in rankings	179,892	121,069
#2: ratio of rankings	176,781	118,487
#3: logarithm of ratio	**174,166**	**116,451**
#4: square root of ratio	174,909	117,051
#5: square of ratio	180,267	118,487

6 Generalized Linear Model Applied to Relationship between ITF Qualifier and WTA Tour Rankings

The ITF serves a series of qualifying rounds for women who intend to play on the top-tier WTA Tour. The preliminary question in this section is how we can predict WTA rankings from ITF rankings for players competing on both in any given year. (The men's ATP has Futures and Challengers circuits as qualifiers, but not discussed herein.)

To start off with the figure on the left, we exhibit the estimated probability of a player in each year having participated in the ITF qualifiers conditional upon their playing a match in the WTA (this is shown in the blue line plot, and since the 1990s over 80% of WTA players have been involved with ITF-level tournaments in the same year; the remainder may be automatically qualified according to possessing enough ranking points in the previous year's WTA). Conversely, the proportion of players in the ITF who qualify for the WTA (shown in red) is a roughly declining function over time from nearly 90% in 1984 down to 20% in the 2020s.





And in the figure on the right, we see the strong connection between ITF and WTA rankings for this common subset of players. It does not necessarily lead to the same implied statistical relationship for all players in general, such as the increasing proportion of ITF contestants who do not qualify for the WTA; and we cannot model this in a straightforward manner due to the issue of missing data.

Two functional forms of the generalized linear model are up for selection, a linear and log-log model.

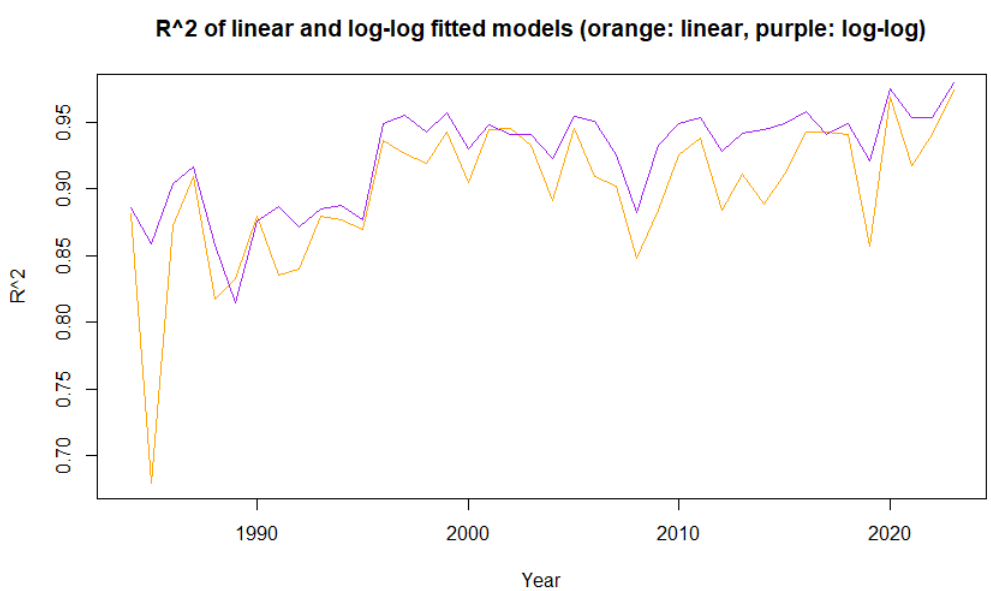
1. Linear:

   $WTA_rank \sim \beta_0 + \beta_1 ITF_rank + \varepsilon$

2. Log-log:

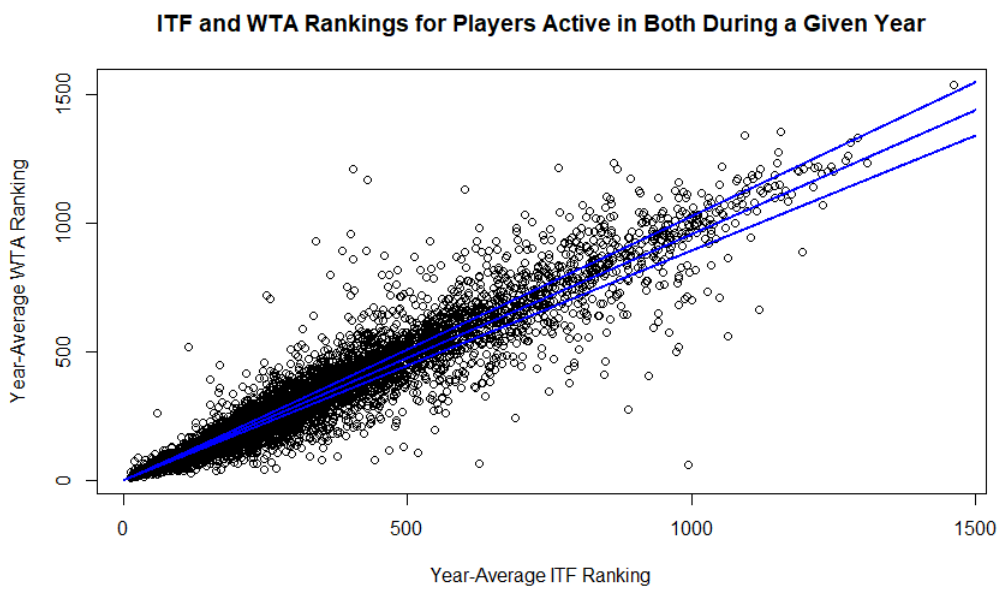
$$\log(WTA_rank) \sim \beta_0 + \beta_1 \log(ITF_rank) + \varepsilon, \text{ or equivalently,}$$
$$WTA_rank \sim e^{\beta_0+\varepsilon}(ITF_rank)^{\beta_1}$$

Each year of data was treated as a subset, and R^2 for each of the two models fitted on that year's data are plotted below; in 90% of years, the log-log model that applied a logarithm on the predictor (feature transformation) and response (link function) demonstrated a higher R^2 than the linear model.



Having established the log-log model as our choice in terms of accuracy, we perform 1000 non-parametric bootstrap samples (with replacement) to provide 95% confidence intervals on the estimated coefficients. The intercept term CI was $[-0.13, -0.07]$ and the slope term CI was $[1.00, 1.01]$. With these bands, we overlay the log-log fit on the scatterplot of ITF versus WTA rankings in order to visualize

the degree of uncertainty in our model. Importantly, note how the log-log transform reflects increasing heteroskedasticity with respect to numerical rankings.



7 Graphical Theory of Domination Cycles

Between any pair of players (u,v) who have played one or more matches against each other, let us define the "domination" of u over v as

$$D(u,v) = P(u,v) - P(v,u) = P(u,v) - [1 - P(u,v)] = 2P(u,v) - 1$$

which can be understood as the difference between the win rate $P(u,v)$, of u against v, and the win rate $P(v,u)$, of v over u. We define $P(u,v) = \frac{a}{a+b}$ when player u has a wins against v, and player v has b wins against u, and hence $D(u,v) = \frac{a-b}{a+b}$. It is straightforward to see that $D(u,v) + D(v,u) = 0$ (zero-sum game).

Let $G(V,E)$ be the directed graph containing V the set of players and $E \subset V \times V$ the set of (directed) arcs where u is adjacent to v whenever $D(u,v) > 0$. In other words, this is a digraph where positive weights represent domination in match win percentages.

A directed cycle C in G is a sequence of vertices $(u_1, \ldots, u_k, u_1)$ where the first k vertices are unique, and for each arc $a \in A = (u_j, u_{(j+1)\%k})$ we have $D(a) > 0$. The total weight $W(C)$ is the sum of its arc weights: $W(C) = \sum_{a \in A} D(a)$, and the concept of average weight can be defined as $\overline{W(C)} = \frac{W(C)}{k}$.

The importance of domination cycles thus defined is that they contradict the consistency of ranking lists. If a higher-ranked player will always win against a lower-ranked player with asymptotic probability 1, then given sufficiently many matches, domination cycles will eventually not exist in our digraph. In the setting of finite samples, the frequency of cycles with large weights decreases as sample size grows.

To illustrate in more familiar terms to some readers, a directed 3-cycle occurs in the game of rock-paper-scissors, where rock beats scissors, and scissors beats paper, but paper beats rock – there is no strict ordering among the three choices. There are no weights in rock-paper-scissors, which is equivalent to the problem in digraphs by assigning $W(a) = 1$ to each arc a. In a generalized version of the game, we want to find the longest cycle: maximize k for which some k-cycle exists in the digraph. This is a known problem that can be solved in linear time and space complexity, $O(|V|)$ [6].

Returning to the weighted digraph, for the moment we restrict our attention to 3-cycles, as $k = 3$ is the minimum cycle length in a digraph defined by our definition of the domination function D; there can be no self-loops, and $D(u,v) > 0$ implies that $D(v,u) < 0$ necessarily so there can be at most one arc between each pair of players. For this relatively small cycle length, it is easy to find 3-cycles of maximum weight by exhaustive search (up to $O(|V|^3)$ for dense graphs), using the following pseudocode – for full details, please refer to the appendix at the end of this submission:

1. For each (unordered) pair of players $\{u,v\}$, if $D(u,v) > 0$ then add an arc in the adjacency list of u toward v, or if $D(u,v) < 0$ then add an arc in the adjacency list of v toward u.

2. For each player u, for each player v in the adjacency list of u, and for each player w in the adjacency list of w, if $D(w,u) > 0$, then we have found a 3-cycle. Compute its weight, keeping track of the triple of players $C' = (u', v', w')$ for the maximum weight $W(C')$ found so far.

3. Output the maximum weight $W(C')$ and the 3-cycle.

Because we want to ensure that the estimated function D is not merely a statistical artifact, we can modify the basic algorithm in one of two ways proposed, which is well-defined for any k, $3 \leq k \leq |V|$, where $|V|$ is the total number of players under consideration:

(a) Hard-thresholding $D(u,v)$ to zero if players u and v did not play a minimum of m matches, where m is a pre-determined constant. The $m = 1$ case is the default algorithm aforementioned.

On the ATP tour for $k = 3$, we have an example where $m = 20$ and the maximum total cycle weight $W(C') = 1.37$ betwixt Ilie Năstase (vertex u), Jimmy Connors (vertex v), and Vitas Gerulaitis (vertex w) who were primarily active throughout the 1970s to 1980s. The corresponding domination arc weights are $D(u,v) = 0.11$, $D(v,w) = 0.44$, and $D(w,u) = 0.82$.

On the WTA tour, we have another famous three-way rivalry betwixt Lindsay Davenport (u), Martina Hingis (v), and Arantxa Sánchez Vicario (w), where the plurality of their matches (≥ 20 between each pair) occurred during the 1990s and 2000s. The maximum total cycle weight is

$$W(C') = 1.09 = 0.12 + 0.82 + 0.17 = D(u,v) + D(v,w) + D(w,u)$$

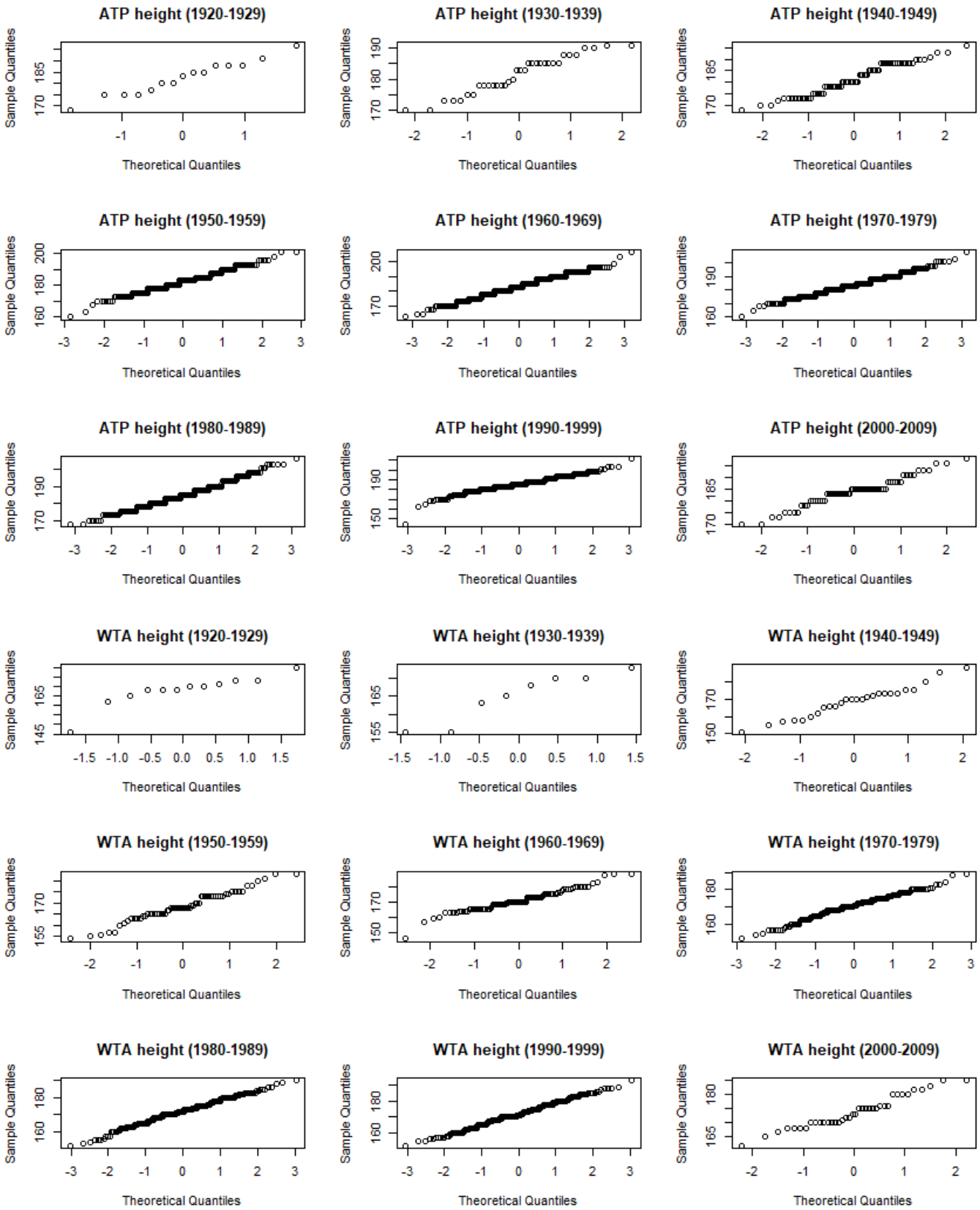
(b) Yet another idea is to replace $D(u,v)$ by the Binomial null hypothesis p-value $p(u,v)$ determined by number of matches won and played. Then the overall weight becomes the log-likelihood

$$L(C) = \sum_{j=1}^{k} \log(p(u_j, u_{(j+1)\%k}))$$

8 Longitudinal Changes in Player Heights

A figure that is often quoted by sports pundits and traditionally displayed on television before the start of a match is the opposing players' heights. Taller players tend to lever faster serves and possess larger wingspans for greater reach, but are less nimble (requiring more time to turn and accelerate, a key athletic requirement in racquet sports) and have greater difficulty retrieving low-skidding shots as is common on the grass courts of Wimbledon; in other words, it is a physiological characteristic that can influence playstyles. How did the variable of height change across the decades?

As a preparatory check, in each decade from the 1920s onward, given a sufficient amount of data, we recognize from theoretical-empirical quantile normal plots that touring professional tennis players' heights followed roughly a normal distribution, similar to the general population (although these athletes are taller on average).


ATP height (1920-1929)
ATP height (1930-1939)
ATP height (1940-1949)
ATP height (1950-1959)
ATP height (1960-1969)
ATP height (1970-1979)
ATP height (1980-1989)
ATP height (1990-1999)
ATP height (2000-2009)
WTA height (1920-1929)
WTA height (1930-1939)
WTA height (1940-1949)
WTA height (1950-1959)
WTA height (1960-1969)
WTA height (1970-1979)
WTA height (1980-1989)
WTA height (1990-1999)
WTA height (2000-2009)
Sample Quantiles
Theoretical Quantiles

Proceeding, we can compare each decade (or colloquially, "generation") of player against the next with one-tailed t-tests (since some samples are smaller, and variance is estimated):

H_0: Mean height of previous generation is the same as the next ($\mu_{[G-10,G)} = \mu_{[G,G+10)}$)

H_1: Next generation is on average taller than the previous generation ($\mu_{[G-10,G)} < \mu_{[G,G+10)}$)

where to be precise $\mu[T_1, T_2)$ is the arithmetic mean of players born between at any time $T_1 \leq t < T_2$, or in plain English, for each generation G in $[1920, \ldots, 2000]$, we compare players born in $[G–10, G)$ versus players born in $[G, G+10)$.

The following table shows p-values for each value of G – note that we test the ATP and WTA players separately; stdev. is abbreviated for *sample* standard deviation and p-values are computed for each t-test of difference in means:

	ATP players		WTA players	
Generation	Mean height (stdev.)	p-value	Mean height (stdev.)	p-value
1920s	182.3cm (7.6cm)	0.7047	167.8cm (8.2cm)	0.1073
1930s	181.4cm (6.1cm)	0.6542	164.9cm (6.8cm)	0.8030
1940s	181.3cm (6.5cm)	0.5486	168.6cm (8.9cm)	0.1139
1950s	182.3cm (6.6cm)	0.1296	168.6cm (6.4cm)	0.5012
1960s	183.1cm (6.3cm)	0.0498	170.8cm (6.7cm)	0.0205
1970s	183.6cm (6.5cm)	0.0596	170.7cm (6.2cm)	0.5651
1980s	184.8cm (6.4cm)	0.0015	172.1cm (6.5cm)	0.0031
1990s	185.3cm (6.7cm)	0.1141	171.9cm (7.0cm)	0.6501
2000s	183.9cm (5.9cm)	0.9599	173.7cm (5.6cm)	0.0315

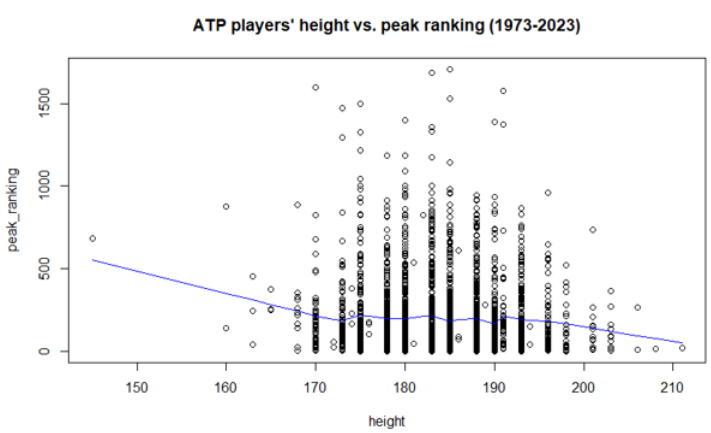
There seems to be certain generational shifts in height as players became taller ($\alpha = .05$), such as in the 1960s and 1980s on the ATP; the 1960s, 1980s, and 2000s on the WTA. To determine why this occurred would demand subject matter research on the history of tennis. As a technical point, we don't calculate family-wise error rates because we have a hypothesis about each specific decade, and we're not testing whether there is at least one generation with an increase in mean height from its predecessor. It is valuable to point out that these values of significance would be discrepant if we grouped players in a different manner than by decade, so the choice of partition here is arbitrary and primarily for ease of interpretation only.

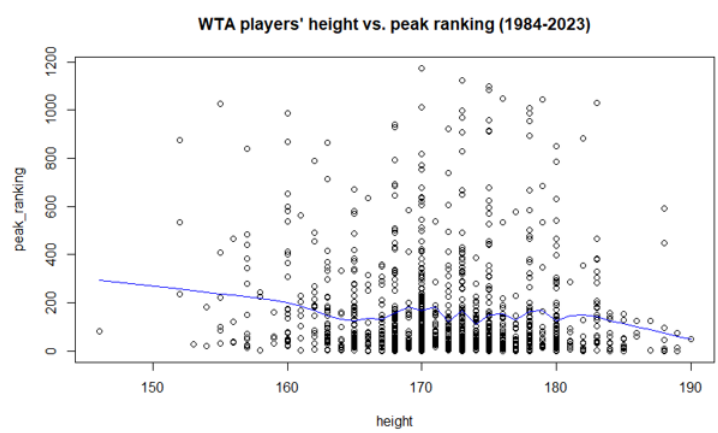
9 Relationship Between Player Heights and Peak Ranking

We now turn to another question on the same variable of interest - is there any association between a player's height and their peak ranking achieved?

Data on rankings are available for the ATP from 1973 until 2023, and on the WTA from 1984 until 2023. From the ranking data which has periodically updated player rankings for each decade, we determine the peak ranking throughout the entire time period. On both tours, the top ranking of #1 was recorded, with a median peak ranking of #138 on the ATP and #74 on the WTA (this median *peak* ranking does not need to be at the halfway point), maximum of #1707 and #1173 respectively. Peak ranking appears to follow a power-law distribution, but we can test this by fitting a discrete power law and bootstrapping, using 100 trial simulations, whence we find that the ATP peak ranking distribution is not significantly different from the theoretical power law fit, while for the WTA they are significantly different (at the .05 level).

It can be difficult to recognize weak correlations from scatterplots alone; however, using Friedman's "SuperSmoother" [3], we can visually identify a negative trend between height and peak ranking for both the ATP and WTA. Although there are men ranked in the top 10 who are as short as 168cm and top 10 women as short as 158cm, in general taller players ranked better than those 170cm or shorter on the men's tour or 165cm on the women's tour, respectively.





Due to outliers, it is tricky to discern whether this pattern is spurious. Another way we can answer the same question is with 95% confidence intervals on the sample geometric mean of peak ranking for each height bucket (and here, we will split into four quartiles for a comfortable bias-variance tradeoff).

ATP heights	Lower bound (95% CI)	Geometric mean	Upper bound (95% CI)
145-180cm (Q1)	110	122	135
180-183cm (Q2)	87	267	1656
183-188cm (Q3)	98	86	112
188-211cm (Q4)	82	92	104

Among the gentlemen, there is a non-monotonic relationship between height and expected peak ranking. Pro male tennis players in the top two quartiles of height seem to have better peak rankings.

WTA heights	Lower bound (95% CI)	Geometric mean	Upper bound (95% CI)
146-168cm (Q1)	69	82	97
168-172cm (Q2)	68	81	97
172-176cm (Q3)	51	63	76
176-190cm (Q4)	41	50	62

The pattern among the ladies is more apparent, with those in the top quartile ranking significantly better than the bottom quartile of height at the peak of their career performance.

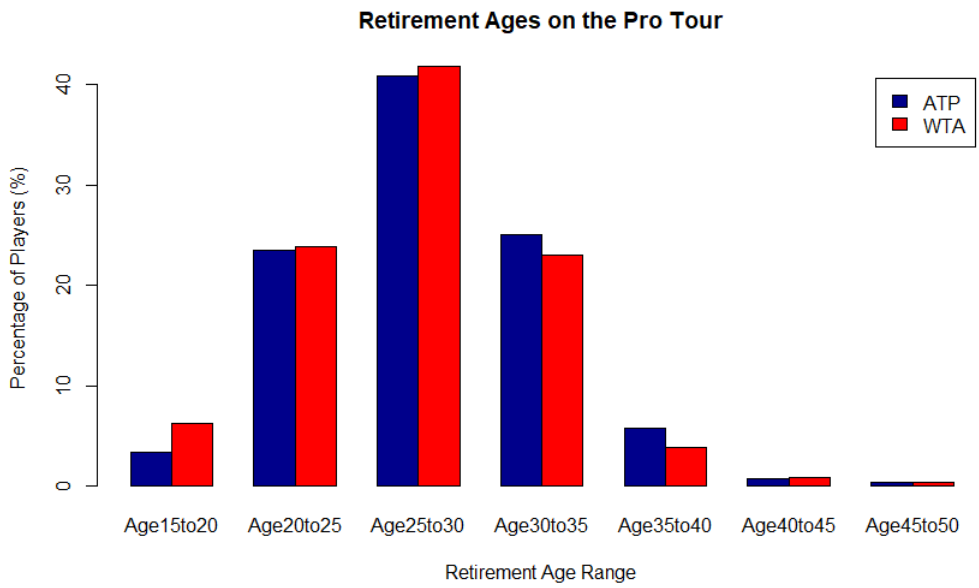
10 Comparison of Univariate Conditional Distributions Using the Histogram Smoother

We have singles match data for both the ATP and the WTA tours from the 1968 to 2013 period (earlier data is also available for the WTA, but let us choose the same time frame to avoid confounding by era). Each match is played on either carpet, clay, grass, or hard court.

From the match data, we can generate empirical distributions for each ATP and WTA player's "dropout" or "retirement" age - these terms will be used interchangeable in semantic context, defined as the player's age when their last match played on their respective tour.

We define retirement by the last match played on the ATP Tour or WTA Tour only, thus excluding from consideration the lower-tiered Challengers Tour, Futures Circuit, and amateur tournaments because the earnings from these series are too usually low to break even on the costs incurred by professional tennis

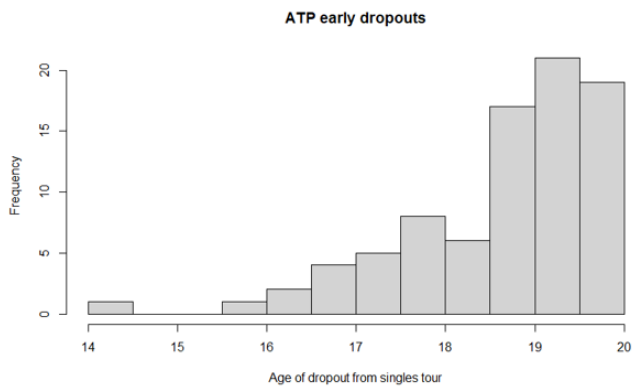
players, not to mention make a sustainable living. Both winning and losing matches are covered, especially because some well-known players before public announcement of their retirement have continued to receive sponsorships and guaranteed appearance fees, for as much as showing up (even if they do not advance far in the rounds). We define the bins in the age histogram by 5-year intervals, starting from age 15 and up to age 50; the intervals are left-closed and right-open, but ages are reported in decimal format so as approximately continuous random variables they are unlikely to be exactly on the integer boundaries.

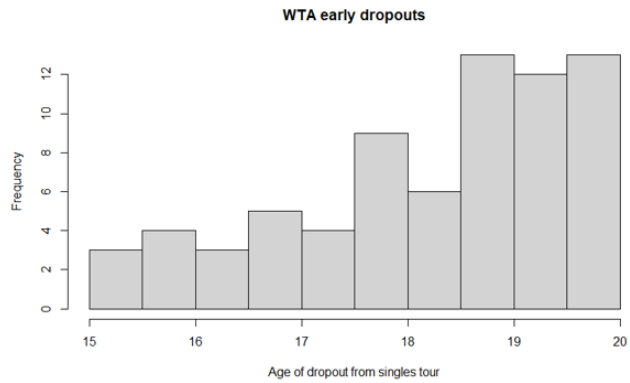


Dropout or retirement age range	ATP proportion (standard error)	WTA proportion (standard error)	ATP - WTA (standard error of difference)
$[15,20)$	$3.4\%(0.4\%)$	$6.2\%(0.7\%)$	$-2.8\%(0.8\%)$
$[20,25)$	$23.5\%(0.9\%)$	$23.9\%(1.3\%)$	$-0.4\%(1.5\%)$
$[25,30)$	$40.9\%(1.0\%)$	$41.8\%(1.4\%)$	$-0.9\%(1.8\%)$
$[30,35)$	$25.1\%(0.9\%)$	$23.0\%(1.2\%)$	$+0.2\%(1.5\%)$
$[35,40)$	$5.8\%(0.5\%)$	$3.8\%(0.6\%)$	$+2.0\%(0.7\%)$
$[40,45)$	$0.7\%(0.2\%)$	$0.8\%(0.3\%)$	$-0.0\%(0.3\%)$
$[45,50)$	$0.4\%(0.1\%)$	$0.4\%(0.2\%)$	$-0.0\%(0.2\%)$

The proportion of players who dropout (presumably before retiring altogether from the workforce) before the age of 20 on the ATP is significantly lower than on the WTA ($z = -3.6$), and the proportion who retire between the age of 35 to 40 on the ATP is significantly higher than on the WTA ($z = 2.8$).

We can zoom in on the conditional distribution of "early" dropouts, namely ATP and WTA players whose last singles match on the tour was played before the age of 20. Observationally, the empirical conditional age distribution of the ATP's early dropouts clearly drops off more quickly than that of the WTA's early dropouts who exhibit a longer tail to the left. Plots are shown separately to distinctly juxtapose the dissimilar shapes of the histograms.





In principle, we can next check if there is a discrepancy between the proportion of ATP and WTA early dropouts from the singles tour who quit to specialize in the doubles format versus the rest of the pro tour, but unfortunately the dataset we have only has ATP doubles matches from 2000 onward, and WTA doubles data is not readily downloadable, so there is limited motivation to pursue this route of inquiry. One hypothesis is that males tend to reach age of peak physical development later in age, whereas relatively more female players are able to compete on tour by their teenage years.

11 Data Mining for Non-Linearity of Partial Correlations in Match-play Data

The purpose of this section is to inspect differences between linear vs. rank-based (nonparametric) partial correlations among a subset of covariates to identify nonlinear relationships. Partial correlations allow us to identify relationships between pairs of random variables when the values of other potential confounders are controlled.

Because naïve win rate estimates can be highly unstable for some players with only a small number of matches recorded, we instead compute the lower bound of the 95% Binomial confidence interval as a

more conservative estimate (which should have less of an effect on players that have a larger sample size of match results).

Additional features engineered are the proportions of a player's matches held on each of the four court surfaces (carpet, clay, grass, or hard), average number of aces per match (points won immediately by service), average number of double faults (points lost by missing both serve attempts in a row) per match, and handedness (encoded as left: +1, ambidextrous or unknown: 0, right: -1).

First, we estimate Pearson's correlation as a measure of linear correlation:

	height	birth_year	retirement_age	win_rate	carpet_proportion	clay_proportion	grass_proportion	hard_proportion	aces	double_faults	handedness
height	1.00	0.06	-0.03	0.00	-0.01	-0.01	-0.01	-0.01	0.44	0.06	0.00
birth_year	0.06	1.00	-0.26	0.11	0.07	0.08	0.08	0.09	0.18	-0.14	0.01
retirement_age	-0.03	-0.26	1.00	-0.40	-0.12	-0.12	-0.12	-0.12	0.21	-0.13	0.04
win_rate	0.00	0.11	-0.40	1.00	-0.08	-0.08	-0.08	-0.08	-0.01	0.04	-0.03
carpet_proportion	-0.01	0.07	-0.12	-0.08	1.00	-1.00	-1.00	-1.00	0.02	0.05	0.01
clay_proportion	-0.01	0.08	-0.12	-0.08	-1.00	1.00	-1.00	-1.00	0.02	0.05	0.01
grass_proportion	-0.01	0.08	-0.12	-0.08	-1.00	-1.00	1.00	-1.00	0.02	0.06	0.01
hard_proportion	-0.01	0.09	-0.12	-0.08	-1.00	-1.00	-1.00	1.00	0.02	0.05	0.01
aces	0.44	0.18	0.21	-0.01	0.02	0.02	0.02	0.02	1.00	0.20	-0.02
double_faults	0.06	-0.14	-0.13	0.04	0.05	0.05	0.06	0.05	0.20	1.00	0.00
handedness	0.00	0.01	0.04	-0.03	0.01	0.01	0.01	0.01	-0.02	0.00	1.00

(ATP Tour: linear correlation coefficients)

Ignoring pathological pairs (such as proportion of matches played on each of the four surfaces, which are linearly dependent), we observe a few noteworthy observations: taller players serve more aces per match ($r = 0.44$ between aces and height), players who retire later have a lower win rate overall ($r = -0.40$ between retirement_age and win_rate), and players who serve more aces also have more double faults ($r = 0.20$, which makes sense if they generally attempt more aggressive and riskier serves).

	height	birth_year	retirement_age	win_rate	carpet_proportion	clay_proportion	grass_proportion	hard_proportion	aces	double_faults	handedness
height	1.00	0.06	0.03	-0.01	-0.06	-0.06	-0.06	-0.06	0.48	0.10	0.02
birth_year	0.06	1.00	-0.47	0.13	0.09	0.11	0.11	0.11	0.11	-0.02	-0.03
retirement_age	0.03	-0.47	1.00	-0.24	-0.01	-0.01	-0.01	-0.01	0.12	-0.17	-0.03
win_rate	-0.01	0.13	-0.24	1.00	-0.05	-0.06	-0.06	-0.06	0.04	-0.09	0.03
carpet_proportion	-0.06	0.09	-0.01	-0.05	1.00	-1.00	-1.00	-1.00	0.04	0.06	0.00
clay_proportion	-0.06	0.11	-0.01	-0.06	-1.00	1.00	-1.00	-1.00	0.04	0.05	0.00
grass_proportion	-0.06	0.11	-0.01	-0.06	-1.00	-1.00	1.00	-1.00	0.04	0.05	0.00
hard_proportion	-0.06	0.11	-0.01	-0.06	-1.00	-1.00	-1.00	1.00	0.04	0.05	0.00
aces	0.48	0.11	0.12	0.04	0.04	0.04	0.04	0.04	1.00	0.03	-0.02
double_faults	0.10	-0.02	-0.17	-0.09	0.06	0.05	0.05	0.05	0.03	1.00	-0.02
handedness	0.02	-0.03	-0.03	0.03	0.00	0.00	0.00	0.00	-0.02	-0.02	1.00

(WTA Tour: linear correlation coefficients)

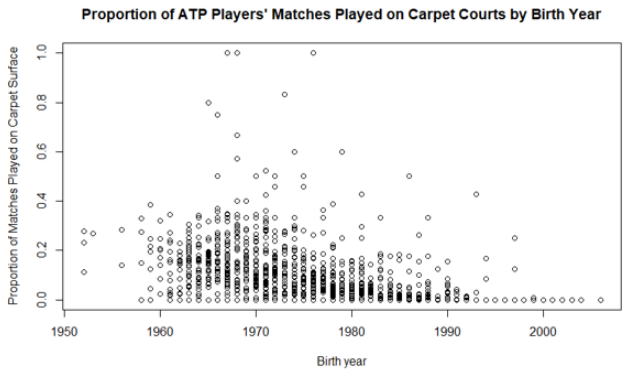
The relationship between average aces per match and height is even stronger on the WTA ($r = 0.48$), and again retirement age is negatively correlated with win rate ($r = -0.24$); this time though, the correlation between the average number of aces and double faults per match is much weaker ($r = 0.03$, not substantially far from zero).

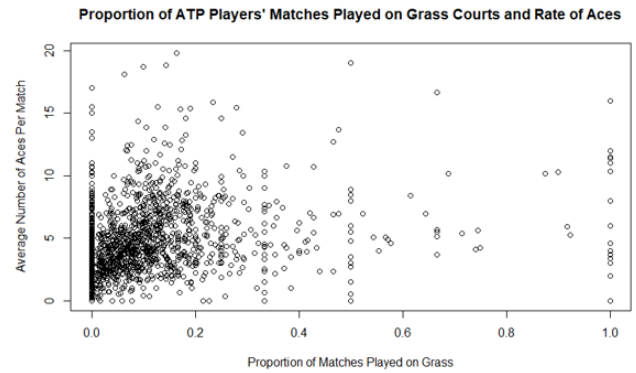
Next, we subtract the matrix of Spearman's rank correlation, ρ, by the corresponding matrix of Pearson's coefficient, r (and since correlation ranges between -1 and 1, each element in the new difference matrix will be a value between -2 and 2); we highlight positive nonlinearity (where the nonlinear correlation is more positive, or less negative, than the linear correlation) in green and negative nonlinearity in red:

	height	birth_year	retirement_age	win_rate	carpet_proportion	clay_proportion	grass_proportion	hard_proportion	aces	double_faults	handedness
height	0.00	-0.03	-0.04	0.06	0.06	0.08	0.05	0.05	-0.05	0.02	0.00
birth_year	-0.03	0.00	0.10	-0.03	-0.59	0.07	-0.04	0.13	0.02	0.06	-0.01
retirement_age	-0.04	0.10	0.00	0.22	0.32	0.42	0.43	0.39	-0.06	0.03	0.00
win_rate	0.06	-0.03	0.22	0.00	-0.07	-0.27	-0.33	-0.22	0.07	-0.10	-0.01
carpet_proportion	0.06	-0.59	0.32	-0.07	0.00	0.76	0.88	0.78	0.05	-0.07	-0.05
clay_proportion	0.08	0.07	0.42	-0.27	0.76	0.00	0.41	0.13	-0.03	-0.22	-0.05
grass_proportion	0.05	-0.04	0.43	-0.33	0.88	0.41	0.00	0.48	0.21	-0.04	0.02
hard_proportion	0.05	0.13	0.39	-0.22	0.78	0.13	0.48	0.00	0.08	-0.18	-0.06
aces	-0.05	0.02	-0.06	0.07	0.05	-0.03	0.21	0.08	0.00	-0.01	0.00
double_faults	0.02	0.06	0.03	-0.10	-0.07	-0.22	-0.04	-0.18	-0.01	0.00	0.00
handedness	0.00	-0.01	0.00	-0.01	-0.05	-0.05	0.02	-0.06	0.00	0.00	0.00

(ATP Tour: nonlinearity difference matrix)

The most evident differences are in the relationship between birth year and proportion of matches played on carpet ($\rho - r = -0.59$, which is nonlinear due to the historical fact that carpet courts became increasingly prevalent during the middle of the 20th century and later fell out of popularity after many professional athletes complained about risk of injuries from slipping on such surfaces), similar time-dependent trends between retirement age and proportion of matches played on disparate court surfaces, along with average number of double faults per match played on clay ($\rho - r = -0.22$), and average number of aces per match played on grass ($\rho - r = 0.21$) for which we show the scatterplot with strong tail dependence between 0% and 20% of matches played on grass that then plateaus beyond the 40% mark (and one possible explanation is that the tennis ball tends to travel faster after bouncing on grass than other surfaces, leading to a higher likelihood of landing an ace, but players with a very high proportion of matches on grass have smaller match sample sizes due to not being able to qualify for as many rounds in carpet, clay, and hard court tournaments so we see diminishing marginal benefit in terms of ace rates).

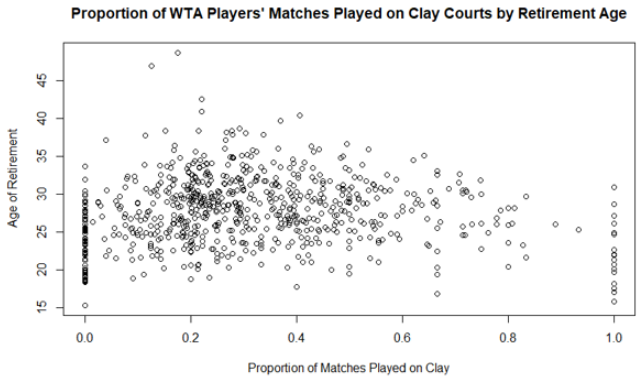


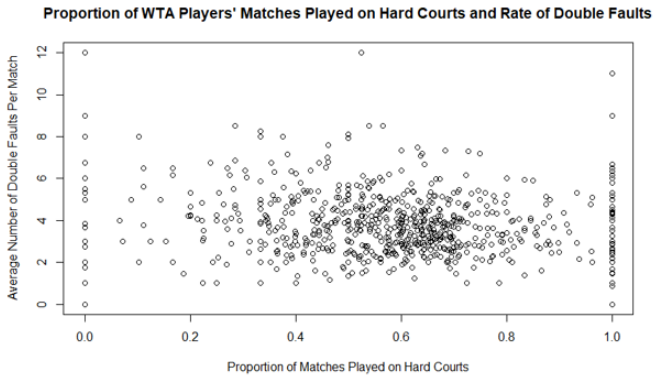


	height	birth_year	retirement_age	win_rate	carpet_proportion	clay_proportion	grass_proportion	hard_proportion	aces	double_faults	handedness
height	0.00	-0.04	-0.03	0.08	0.09	0.18	0.09	0.14	-0.05	0.01	0.02
birth_year	-0.04	0.00	0.11	0.03	-0.72	0.12	0.02	0.17	0.03	0.00	-0.01
retirement_age	-0.03	0.11	0.00	0.13	0.04	0.25	0.24	0.22	-0.05	0.01	0.00
win_rate	0.08	0.03	0.13	0.00	0.11	-0.26	-0.26	-0.24	0.05	-0.02	-0.04
carpet_proportion	0.09	-0.72	0.04	0.11	0.00	1.07	1.13	1.04	0.04	-0.05	-0.03
clay_proportion	0.18	0.12	0.25	-0.26	1.07	0.00	0.39	0.13	-0.01	-0.13	-0.02
grass_proportion	0.09	0.02	0.24	-0.26	1.13	0.39	0.00	0.45	0.24	-0.07	-0.06
hard_proportion	0.14	0.17	0.22	-0.24	1.04	0.13	0.45	0.00	0.06	-0.17	-0.06
aces	-0.05	0.03	-0.05	0.05	0.04	-0.01	0.24	0.06	0.00	0.05	0.01
double_faults	0.01	0.00	0.01	-0.02	-0.05	-0.13	-0.07	-0.17	0.05	0.00	-0.03
handedness	0.02	-0.01	0.00	-0.04	-0.03	-0.02	-0.06	-0.06	0.01	-0.03	0.00

(WTA Tour: nonlinearity difference matrix)

Similar signs of nonlinearity can be seen among WTA players. In particular, there is positive nonlinearity between proportion of matches played on clay and retirement age ($\rho - r = 0.25$), and the relatively more negative nonlinear correlation between proportion of matches played on hard courts versus rate of double faults ($\rho - r = -0.17$) due to the robustness of Spearman's nonparametric rank-based method that is less sensitive to outliers.





An alternative approach which we could have taken in this presentation would be a difference in t-statistics (since they are on the same scale) rather than correlations; or Kendall's tau could be used instead because it is a monotonic function of Spearman's ρ and will yield the same p-values. On the other hand, t-statistics are unbounded and thus less amenable to interpretation than a standardized coefficient.

12 Future Work

The following questions would be of interest for further investigation, but are omitted from this project due to limitations on length and scope:

- How accurately does the approximation of equal service and return win rates affect predicted set and match win rates as random functions of game win rate?
- Does playing surface play a meaningful role into players' optimal height (as determined by expected peak ranking)? Usually, tennis balls bounce lower on grass courts and higher on clay courts (which makes the point of racquet contact more comfortable for taller players).
- Is a player's favored court surface (in terms of their highest match win rate, relative rate of matches played, or best tournament results such as number of finals reached) a predictor of injury rate and retirement age?
- What is the theoretical probability of finding beyond a certain number of k-cycles of minimum weight W in a domination digraph of N players and M matches played across T randomly seeded tournaments?
- Are there any identifiable common causes for early retirement from the singles tour – perhaps switching to a career in doubles tennis (or other sports)?
- Do match durations follow a Poisson distribution? How about other variables of time, such as players' career lengths?
- Extend the ITF qualifier and WTA rankings analysis to the ATP system's three-tiered ladder from Futures, to Challengers, and Tour-level events.
- Can we conversely predict the proportion of points that a player has won based on their set and match win rates? As a slight subtlety, due to the optimization problems' loss-minimizing objective functions, this is seldom quite so simple as inverting the fitted regression models.
- Questions that come to mind regarding cliques defined in (undirected) graphs of players who have had a match between each other is the order of the maximum clique, average percentile rank of cliques for each tournament year or decade, and whether there is a difference between these on the ATP and WTA tours; these may provide a lens into the interaction dynamic between top players

who are more likely to be meet each other in persistent rivalries toward later (semi-final or final) tournament rounds.

- Additional more detailed information about players such as their body weight, scores on fitness tests, level of education, mental and physical health diagnoses, racquets and grips, average and peak serve speeds, would be more conducive toward a richer predictive model of match results.

13 Conclusion

Within the hierarchical scoring system, we constructed a discrete-time stochastic process and calculated the probability of reaching each state, with the overall probability expressed as a sum of several polynomial terms and geometric series. A logit transform on probability of winning a game was fit to a high-order polynomial regression model on the probability of winning each point. An analogous problem on the higher layers of scoring in sets and matches was handled by numerical simulation.

Using 2D heatmaps, we visualized a diverse range of match-winning likelihoods between players across ranking deciles, where the deciles are weighted by frequency of matches. And by Akaike's Information Criterion, the logarithm of ranking ratios between the two opponents was shown to be a preferred feature for model accuracy over other functions such as the difference in rankings.

A view of conditional probabilities in qualifying for the WTA via the ITF, and vice-versa the proportion of WTA players who had participated in the ITF tournaments, showed increasing competition over time. These rankings between the ITF and WTA are closely related, although a log-log GLM form was superior to the simple linear regression model across most decades of our data. A scatterplot with 95% CI shows the capture of most datapoint within the estimated bands.

We defined the notion of "domination" in digraphs to formalize the intuition that tennis matches can be viewed as zero-sum games, and outlined ways of defining a weight function on cycles as the aggregation of directed arc weights. Both the ATP and WTA Tour had cycles of order three as found by a graph algorithm with pruning akin to depth-first search.

Normality of physical height distributions (conditional on being a professional tennis player) was validated for the men and women separately, and a generation-by-generation t-test of differences yielded some

evidence of overall toward taller players at the professional level. By leveraging smoothers, we delved into more granular detail of peak player ranking as it relates to their stature. There is less evidence supporting monotonicity for the ATP men, whereas on the WTA, height quartiles differed significantly in the central tendency of players' peak rankings achieved favoring taller players on the women's tour.

An aspect of the WTA is that there are more early dropouts and fewer players whose retirement is more advanced in age relative to that of the ATP, which was evinced through a side-by-side histogram plot.

Finally, we used both linear and non-linear correlations together to generate difference matrices that revealed which pairwise relationships were greater in degree of nonlinearity. Aside, there was confirmation of the positive association between number of aces per match and number of double faults, in addition to the server's height. Players who continued into older age brackets faced lower career-average rankings than those who retired when they were younger. More nonlinearity in pairwise relationships was evident between time-dependent variables such as birth year (hence historical era) and retirement age versus court surface types, which are in turn statistically related to the rate of aces and double faults per match potentially as a reflection of physical trajectory caused by the ball bounce on distinct ground materials.

The predominance of our study is rooted in non-causal inference, although it can be extended to experimental methods of analysis that goes beyond the author's selected scope owing to the most readily available data being observational in nature. Also, we cannot answer every remaining question on the list for future work, but a bibliography and appendix containing mostly R and C++ scripts is shared for extension.

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